

ADVANCED COGNITIVE NEUROSCIENCE

SOURCE RECONSTRUCTION USING MNE

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WHERE ARE WE AT AND WHERE ARE WE GOING?

Provided but you'll need to update your params.py file to be able to run it

- 01_anatomy.py
- 02_forward.py
- 03_filter.py
- 04_ica.py
- 05_epochs.py
- 06_stcs.py

Common scripts for
everyone

I suggest you to have these ready **before week 44**

Coming soon to a GitHub repo near you:

- 07_evoked.py
- 08_erf_stats.py

Inspiration



WHERE ARE WE AT AND WHERE ARE WE GOING?

Before week 44

- Update *params.py* have with:
 - Dates and bad channels for all subjects
 - File paths
 - Relevant analysis parameters
- Scripts 1-6 have been run:
 - Preprocessed epochs, including metadata, are available for each subject
 - Source time courses have been extracted for all targets
- Quality control has been completed or is ongoing:
 - The reports have been looked through



WHERE ARE WE AT AND WHERE ARE WE GOING?

-  Work on your reports
-  Introduction to relevant material for the report
-  Inspiration

Week 37:

Monday: Workshop paradigm: Measuring visual subjective experience
MR Recordings (Lau)

Week 38:

Monday: What is it all about? (Lau)
Workshop 1: MR-recordings (Laura & Lau)

Week 39:

 Monday: Basic physiology and Evoked responses (Lau)
Wednesday: Preprocessing and evoked responses (Laura)

Week 40:

 Monday: Forward modelling and inverse modelling (Lau)
Wednesday: Source reconstruction (Laura)

Week 41:

 Monday: Multivariate statistics (Laura) (Lecture)
Wednesday: Multivariate statistics (Laura) (Coding)

Deadline for feedback: Video Explainer

Week 43:

 Monday: Inverse modelling: beamforming (Lau) (Lecture)
Wednesday: Inverse modelling: beamforming (Laura) (Coding)

Week 44:

 Monday: From single subject analysis to group analysis (Laura)
Wednesday: Work on your reports (Laura)

Week 45:

 Monday: What about that other cortex? - the cerebellar one (Lau)
Wednesday: Work on your reports, continued (Laura)

Week 46:

 Monday: Optically pumped magnetoencephalography (Laura)
Wednesday: Respiration (Laura)

Deadline for feedback: Lab report

Week 47:

No teaching: prepare presentations!

Week 48:

Monday: Oral presentations (part 1) (Lau)
Wednesday: Oral presentations (part 2) (Lau)

Week 49:

Monday: Oral presentations (part 3) (Lau)
Wednesday: Oral presentations (part 4) (Lau)

Week 50:

Wednesday: What was it all about (Lau)

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TODAY

Focus on **content** not so much on **bookkeeping**

Aim: Learn how to do source reconstruction using minimum norm estimation

Many studies use **surface-based source spaces**, but today we'll work primarily with a **volumetric source space**.

- The interactive plots we use to explore source time courses from surface source spaces do not work on uCloud
- The volumetric source space allows us to use interactive visualisations in the notebook, making it easier to explore and understand what the source reconstruction is doing.

To plot a **surface source time course**, you can:

1. Save your source time courses after morphing them to **fsaverage**.
2. Download the source time courses together with the relevant FreeSurfer files for fsaverage.
3. Plot and explore the surface source estimates locally.



LETS GO TO UCLOUD

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Work through the source_recon.ipynb notebook! Do get a

— — — EXTRA — — —

- Start working on the scripts! The code in today's notebook should help you with
 - 01_anatomy.py
 - 02_forward.py
 - 06_stcs.py
- You can work on scripts from previous classes instead if you prefer

Provided but you'll need to update your params.py file to be able to run it

